



Retail Management System

Project Proposal



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Overview

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Introduction



Context

Rapid growth of digital technology has transformed retail into online platforms



Project

Development of a Web-Based Retail Management System with modular system architecture and ML integration



Learning Goals

Demonstrates integration of web development, DB management, software engineering, and machine learning.



Problem Statement

1 Limited Visual Product Search

Most small retail platforms depend only on text-based search making it difficult for users to find products.

2 Lack of Centralized Retail Management

Small businesses often manage products, orders, and customer data manually or through basic systems, leading to inefficiency.

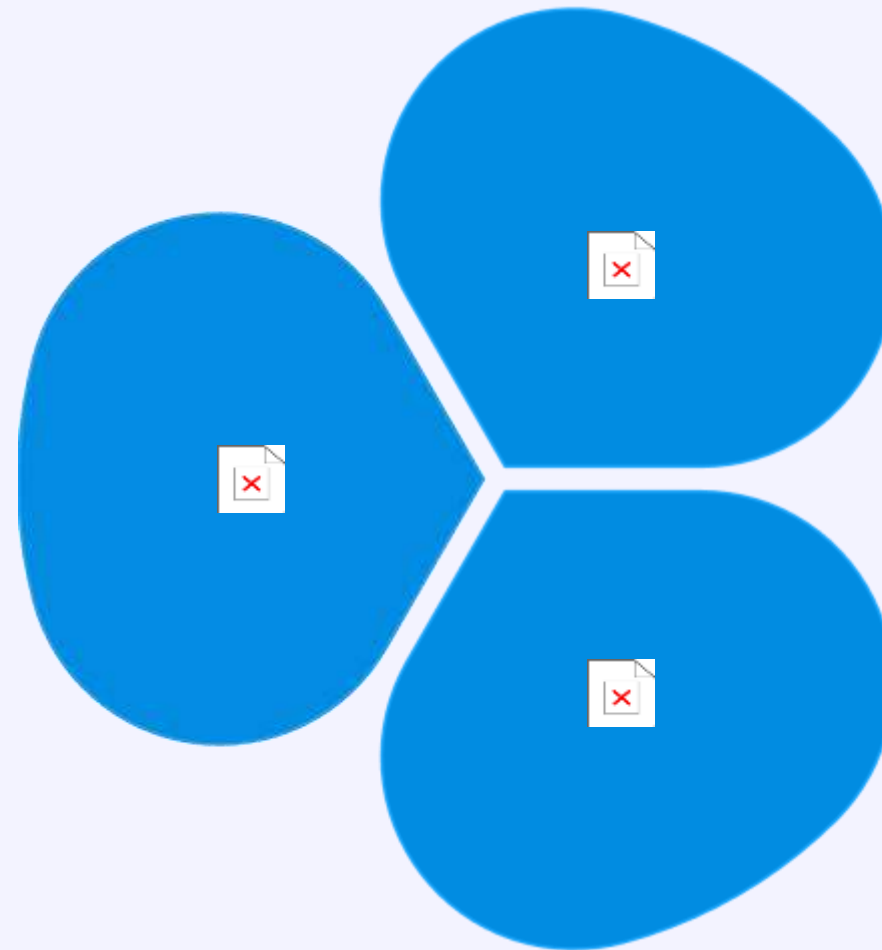
3 Need for a Simple, Scalable Web-Based Solution

There is a need for a structured, secure, and easy-to-manage retail system suitable for small and growing businesses.

Objectives

Centralized Platform

To provide a unified web platform for product browsing and purchasing



Visual Search

To allow users to upload product images for searching

Skill Enhancement & Technical Growth

To gain practical experience in full-stack web development and ML integration

Development Methodology

The project uses the Agile Development Model, which focuses on making progress step by step, getting regular feedback, and easily adjusting to changing needs.



Fig: Agile Development Model [3]

Functional Requirements



User & Access Management

- User registration, secure login profile management
- Role-based permissions for Admin and Customer
- Secure authentication and permission enforcement



Product & Order Management

- Efficient product catalog, search & browse
- Add-to-cart, checkout, order placement & tracking
- Admin panel for product CRUD, orders & inventory overview



Payment & Notification System

- Secure online payment and transaction reliability
- Instant order confirmations and email notifications

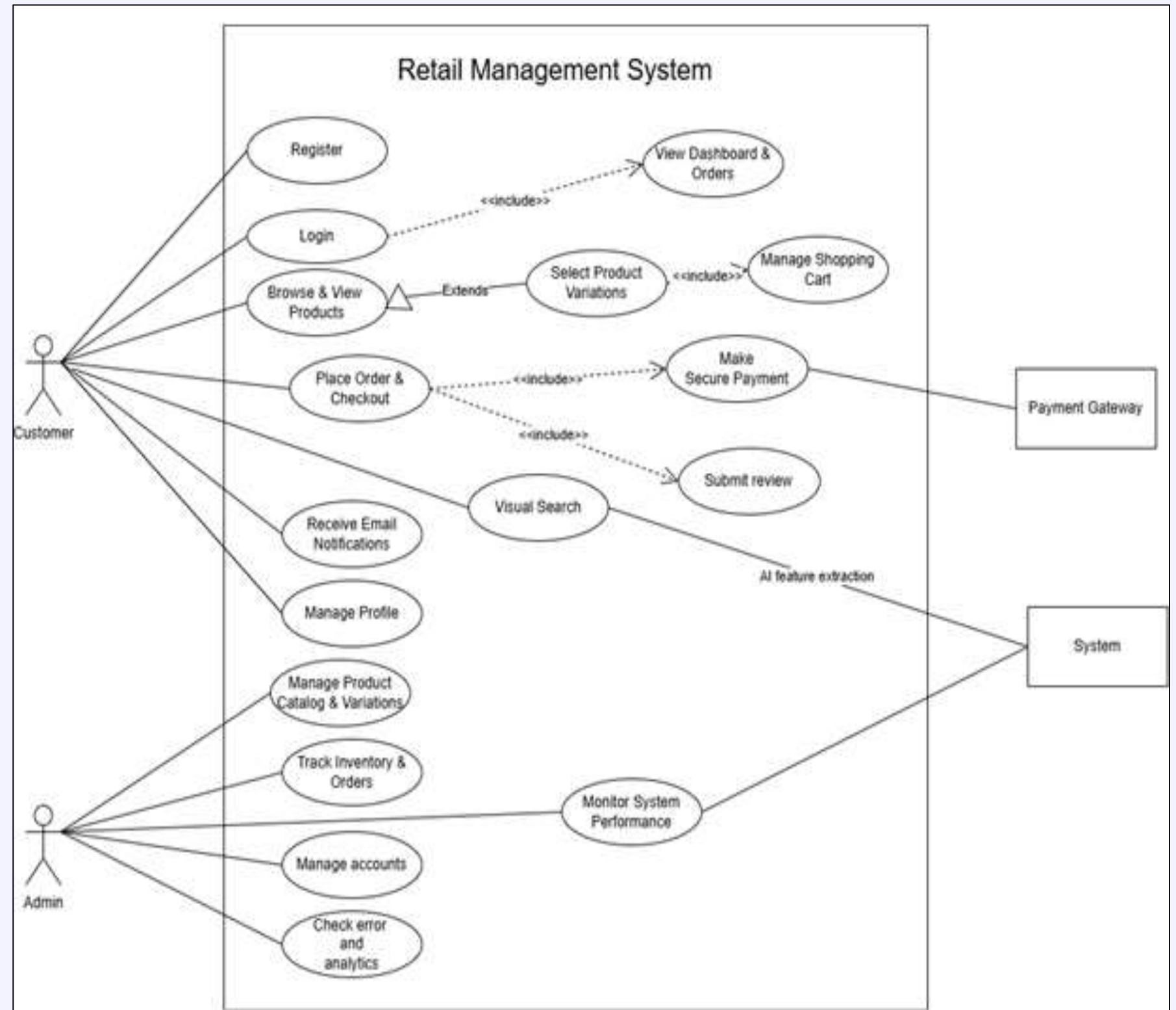


Intelligent Product Search

- Upload images to search by visual similarity
- AI-driven feature extraction and matching

Use Case Diagram

Retail Management System



Non-Functional Requirements



Usability & Compatibility

- User-friendly
- Easy to use without special training
- Accessible on different screen sizes



Performance & Reliability

- Fast response time during browsing and transactions
- Smooth user experience
- Maintains data integrity under unexpected conditions



Security

- Secure authentication and authorization
- Protection of user and transaction data
- Safe data handling mechanisms



Scalability & Maintainability

- Supports future growth in users and products
- Handles increasing transactions efficiently
- Flexible for future enhancements

Schedule Feasibility



Key activities	Duration (in weeks)														
	1st	2nd	3rd	4th	5th	6th	7th	8th	9th	10th	11th	12th	13th	14th	15th
Planning	■	■													
Analysis & design		■	■	■											
Implementation			■	■	■	■	■	■	■	■	■	■	■	■	
Testing													■	■	
Evaluation												■	■	■	
Documentation				■	■	■	■	■	■	■	■	■	■	■	■
Deployment															■



Technical Feasibility



Frontend

- ❖ HTML, CSS, JavaScript
- ❖ jQuery for dynamic interactions
- ❖ Bootstrap for responsive design



Backend

- ❖ Python & Django for the web framework



Database

- ❖ PostgreSQL for reliable data management



AI / ML

- ❖ Python(TensorFlow/Keras)
- ❖ Pre-trained ResNet50 model for feature extraction

Core Algorithms

1. Similarity Search Algorithm (Nearest Neighbors)

- Compare user image vector with all stored vectors
- Use Euclidean Distance to find closest matches.
- Mathematical Logic :

$$d(p, q) = \sqrt{\sum_{i=1}^n (p_i - q_i)^2}$$

2. Feature Extraction

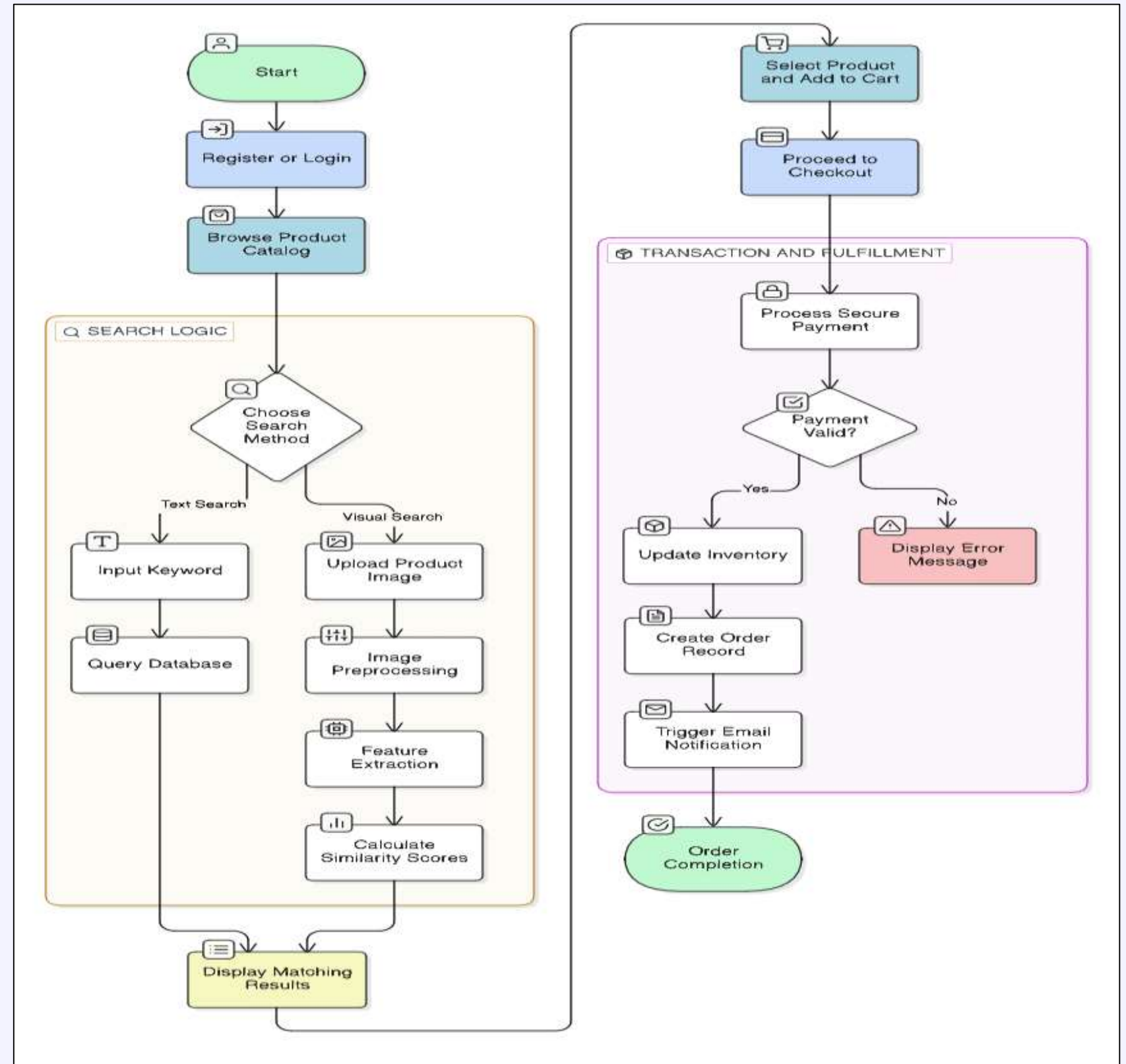
- Pre-trained **ResNet50 CNN** is used
 - Each image converted into a **2048-dimensional feature vector**
-

3. Review Authorization

- Ensures only genuine buyers can submit product reviews
- Prevents fake ratings and review manipulation
- Mathematical Logic :
 - Let:
 - U = User
 - C = Number of completed purchases by user for product
 - Allow Review = {1, if C ≥ 1
0, if C = 0

Flowchart

Retail Management System



Expected Outcome



Improved Discovery

AI-powered visual search reduces dependence on text and boosts conversions.



Scalable Design

Modular, extensible system enabling future enhancements with minimal effort.



Academic Goals

Meets final-year requirements: problem analysis, system design, algorithm integration, feasibility evaluation.

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Thank You!

We appreciate your time and attention.

Any questions?